Heaven's Light is our Guide
DIRECTOR OF RESEARCH & EXTENSION
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Rajshahi-6204, Bangladesh

Application form (Guidelines) for approval of Ph.D/M.Phil/M.Sc.Engg./M.Engg. Thesis/project proposal by the CASR. All items applicable to the following list must be mentioned and filled in correctly. **Please submit twenty-two (22) hard copies** [One original and the other nineteen (21) photocopies in the **entire script plain paper**] and also submit a scanned copy of the proposal.

Date: 15th November, 2025

1. Name of the student: MD. RUHUL AMIN **Status:** Part-Time
(Block Letters)

Roll No.: 2204103014 **Session:** 2022-23

2. Present Address: 206-Chowdhury Palace, Sector-2, Uposhohor, Rajshahi

3. Present Designation & Organization: Software Engineer, Business Automation Ltd.

4. Name of the Department: Computer Science and Engineering **Programme**: MSc Engineering

5. Name of the Supervisor: Dr. Md. Rabiul Islam **Designation**: Professor

6. Name of the Co-Supervisor (if any): N/A **Designation**: N/A

7. Date of First Enrolment in the Programme: March, 2023

8. Tentative Title (Block Letters, Times New Roman, Font size 11 points, Not more than 25 words)**:**
DEEP LEARNING–BASED DETECTION OF COLORECTAL POLYPS USING TR-SE-NET AND ENSEMBLE ELM ARCHITECTURE WITH EXPLAINABLE AI.

9. Background and present state of the problem

(Not more than 250 words. Please mention only those activities carried out in different places as reported in publications. Please support your information by citing the relevant references.)

Colorectal cancer (CRC) remains one of the leading causes of cancer-related mortality worldwide, and early detection of colorectal polyps plays a crucial role in prevention and improved patient outcomes. Traditional colonoscopy, although widely used, relies heavily on manual visual inspection, making it susceptible to intra-observer variability, fatigue, and missed lesions. These challenges underline the necessity for accurate, automated, and interpretable computer-aided detection systems.

Recent advancements in deep learning have significantly contributed to addressing these limitations. Ensemble Extreme Learning Machines (ELM) integrated with Convolutional Neural Networks (CNNs) have shown improvements in detection performance and model interpretability [1]. Transformer-based architectures capture both local and global contextual features, enabling more robust feature extraction [2], while ensemble ELM approaches enhance classification reliability through multi-model aggregation [3]. SE modules have demonstrated the ability to emphasize channel-wise feature importance, boosting segmentation accuracy [4], and CNNs combined with SENet further refine polyp boundary detection [5]. Studies consistently report the drawbacks of manual colonoscopy, including variability among clinicians and frequent oversight of small or flat polyps [6].

Ensemble learning techniques continue to improve generalization and stability in diverse clinical environments [7]. Meanwhile, explainable AI (XAI) methods—such as heatmaps and saliency-based

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visualizations—offer greater transparency and support clinical trust in automated predictions [8]. Additional research reinforces the need for reliable automated systems to complement traditional colonoscopy, particularly in high-volume screening settings [9].

Together, these advancements address challenges of data scarcity, computational efficiency, and interpretability, moving toward reliable, automated, and explainable systems that support early CRC diagnosis and patient care.

10. Justification of the study

(Not more than 150 words. Please justify research/study based on the present state of the problem and works that have been carried out in different places as reported in publications. Please support your information by citing the relevant references.)

The main goal of segmentation-based methods for polyp detection is to identify colorectal polyps in real time, which has not yet been adequately explored in research. Existing automated methods still struggles with segmentation precision, efficient feature extraction, and model generalization (M. F. Ahamed et al. [1]). Although CNN and transformer-based models have achieved progress, the application of the TR-SE-Net model for real-time segmentation-based polyp classification has not yet been explored. In this work, segmentation serves as the essential step to extract the most discriminative polyp features, which are then processed through a PD-CNN and Ensemble Extreme Learning Machine (EELM) architecture to achieve improved classification accuracy and robustness. The integration of TR-SE-Net with PD-CNN and EELM enhances both real-time detection and interpretability. Additionally, Explainable AI (XAI) techniques are incorporated to ensure model transparency and clinical reliability, supporting gastroenterologists with accurate and trustworthy automated polyp detection systems.

11. Objectives with specific aims

(Please list the objectives(s) using short sentences. If you are writing one or two paragraphs describing the objectives, please limit yourself to 150 words.)

This study proposes a deep learning–based system for automated colorectal polyp detection using TR-SE-Net integrated with PD-CNN and ensemble Extreme Learning Machine (ELM) architecture. The main objectives are:

1. To develop an efficient real-time segmentation-based preprocessing and feature extraction pipeline using the TR-SE-Net model for accurate and discriminative polyp feature representation.
2. To design and implement a TR-SE-Net-PD-CNN–Ensemble ELM (EELM) framework that seamlessly integrates deep learning and ensemble learning methodologies to enhance the accuracy, robustness, and interpretability of real-time colorectal polyp classification.
3. To incorporate Explainable AI (XAI) techniques to provide transparent and interpretable explanations for detection and classification decisions, fostering clinical trust.

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12. Outline of Methodology/ Experimental Design

(Outline the approach and the sequence of activities in not more than 500 words to describe how the work will be carried out) ***Figure/Table will be considered as 100 words.)***

An overview of the proposed methodology is shown in Fig. 1. This study aims to develop an automated colorectal disease detection system capable of performing both segmentation and classification for polyps, tumors, and ulcers by integrating TR-SE-Net with an Ensemble Extreme Learning Machine (EELM) framework enhanced through Explainable AI (XAI) techniques. The proposed TR-SE-Net–based segmentation model represents a key contribution of this research, specifically designed to achieve accurate and efficient lesion segmentation in real-time medical imaging applications.

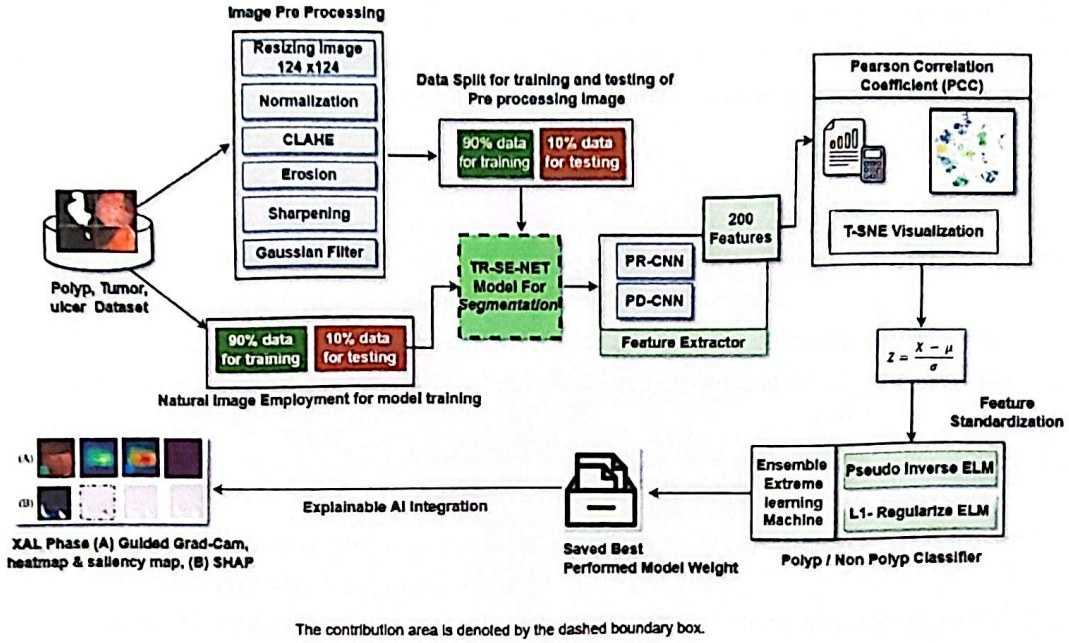


Fig. 1: Proposed Methodology for Deep Learning-Based Detection of Colorectal Polyps Using TR-SE-NET And Ensemble ELM Architecture.

Segmented colonoscopy image datasets containing polyp, tumor, and ulcer samples—such as Kvasir [10]—are utilized for supervised learning. All images undergo preprocessing operations, including resizing, intensity normalization, histogram equalization, and gaussian filter, to standardize input data and improve feature extraction accuracy. Both preprocessed and raw images are then fed into the proposed TR-SE-Net model for precise segmentation. TR-SE-Net employs a hybrid encoder–decoder structure that combines Transformer encoder blocks with Squeeze-and-Excite (SE) modules to enhance channel-wise attention and improve segmentation precision. Skip connections are applied to preserve spatial information between encoder and decoder layers, while ground-truth masks guide optimization using Dice score and Intersection over Union (IoU) metrics.

After segmentation, the extracted feature maps are forwarded to PD-CNN and PR-CNN models for deep feature extraction, producing 200 significant features that represent the most discriminative characteristics of each class.

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These features are further analyzed using the Pearson Correlation Coefficient (PCC) to evaluate inter-feature relationships and t-Distributed Stochastic Neighbor Embedding (t-SNE) for visualizing feature separability in reduced-dimensional space. Feature Standardization is then applied to normalize the extracted features for optimal learning performance.

The standardized features are processed by an Ensemble Extreme Learning Machine (EELM) architecture consisting of Pseudo-Inverse ELM and L1-Regularized ELM classifiers. This hybrid ensemble approach enhances model robustness, reduces overfitting, and improves classification accuracy across polyp, tumor, and ulcer categories. Each classifier independently contributes to the decision process, and their aggregated output enhances generalization and reliability [3].

Finally, to ensure clinical interpretability, XAI techniques such as Heatmaps, Saliency Maps, SHAP, and Guided Grad-CAM are incorporated to highlight the most influential image regions affecting classification decisions. The complete pipeline is evaluated using accuracy, sensitivity, specificity, F1-score, and inference time, ensuring a reliable, interpretable, and clinically applicable real-time system for gastrointestinal disease diagnosis.

13. Expected outcomes

(Please list the possible outcomes using short sentences. If you are writing one or two paragraphs describing the expected outcomes, please limit yourself to 150 words)

In this work, we aim to propose an effective deep learning approach for colorectal polyp detection and classification integrating TR-SE-Net with ensemble ELM and explainable AI. So, expected outcomes include:

a. High-accuracy models for colorectal polyp detection with segmentation and classification capabilities.
b. Models with enhanced robustness and generalization capacity to perform reliably across diverse patient datasets and clinical scenarios.
c. Explainable AI-driven transparent visualizations of decision-making processes to build clinical trust and improve usability in medical practice.

These outcomes collectively aim to improve early diagnosis accuracy, reduce false positives /negatives, and advance automated medical imaging for gastrointestinal disease diagnosis.

14. References

(All the references should be cited inside the texts, and a maximum of 10 references should be used. While giving the references, you must mention the author, title of the paper, and name of the journal /proceeding. Vol. no., Year of publication, etc., in the standard style given below)

[1] M. F. Ahamed, M. Nahiduzzaman, M. R. Islam, et al., "Detection of various gastrointestinal tract diseases through a deep learning method with ensemble ELM and explainable AI," *Expert Systems with Applications*, vol. 256, p. 124908, Dec. 2024.

[2] S. Kumar, et al., "Real-time colorectal polyp detection with transformer-based deep learning architectures," *Scientific Reports*, vol. 14, 2024.

[3] M. F. Ahamed, M. R. Islam, M. Nahiduzzaman, M. J. Karim, M. A. Ayari, and A. Khandakar, "Automated Detection of Colorectal Polyp Utilizing Deep Learning Methods with Explainable AI," *IEEE Access*, vol. 12, 2024.

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[4] L. Zhou, et al., "Interpretability of deep learning models in colonoscopy analysis," *IEEE Transactions on Medical Imaging*, vol. 43, no. 1, 2024.
[5] J. Smith, L. Zhou, R. Wang, and T. Chen, "Automated colorectal polyp detection combining CNN and SE-Net," *Medical Image Analysis*, vol. 78, 2023.
[6] Y. Zhang, Y. Li, Z. Wang, et al., "Deep learning driven colorectal lesion detection in colonoscopy: A systematic review," *International Journal of Molecular Sciences*, vol. 24, no. 5, 2023.
[7] J. Lee, S. Kim, et al., "Ensemble Learning with Extreme Learning Machines for Colorectal Polyp Classification," *Neural Computing and Applications*, vol. 35, no. 7, 2023.
[8] H. Wu, et al., "Explainable AI for medical imaging: Applications in colorectal polyp detection," *Frontiers in Medicine*, vol. 9, 2022.
[9] G. Bernal, F. Sánchez, et al., "Comparative assessment of polyp detection methods in colonoscopy videos," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 2, 2019.
[10] M. Nagadia, "Kvasir dataset," *Kaggle*, 2021. [Online]. Available: https://www.kaggle.com/datasets/meetnagadia/kvasir-dataset. [Accessed: 3-May-2025].

15. List of courses taken (Completed/registered) so far *(To be verified and signed by the Course Advisor, and a Photocopy of Grade Sheets should be provided)*

Sl. No.	Course No.	Course Name	Grade	Credit	Grade point.	G. P. A
1	CSE 6310	Data Mining and Warehousing	A	3.00	3.75	3.75
2	CSE 6501	Advanced Digital Image Processing	A	3.00	3.75	
3	CSE 6701	Advanced Database Management System	A	3.00	3.75	
4	CSE 6106	Combinatorial Optimization	A	3.00	3.75	
5	CSE 6303	Machine Learning	A	3.00	3.75	
6	CSE 6602	Advanced Computer Network	A	3.00	3.75	
			Total	18.00	22.5	

Signature of the Course Advisor: .....

16. Cost Estimate: (Invoice / Quotation must be provided for each item which costs Tk. 5,000/- and more)

(a) Cost of Material	Tk. 2,500 /-
(b) Fieldworks (if applicable)	Tk. N/A
(c) Conveyance/ Data Collection (With Breakup)	Tk. N/A
(d) Drafting, Binding & Paper, etc. (Max. 3000/-)	Tk. 3,000 /-
(e) Miscellaneous	Tk. 2,000 /-
Total:	Tk. 7,500 /-

(Break-ups may be provided in a separate sheet if necessary)

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17. Justification of having Co-Supervisor:

Having a co-supervisor is generally not recommended for a master's thesis unless it is necessary. At the Ph.D. level, co-supervisors may be appointed if needed. In both cases, it must be justified to the main supervisor that the research requires substantial expertise in a field outside their own area of specialization.

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18. Doctoral Committee/PGAC reference:

Meeting no.: ..... **Resolution No.:** ..... **Date:** .....

19. Appointment of Supervisor & Co-Supervisor Approved by the CASR Meeting No. (For Ph. D):-

Resolution No.:..... **Date:**.....

20. Appointment of Doctoral Committee Approved by the CASR Meeting No. (For Ph. D):.....

Resolution No.: ..... **Date:** .....

21. Result of the comprehensive examination for Ph. D (Photocopy of the result should be enclosed)

Date: ..... **Satisfactory/Unsatisfactory.**

22. Number of Post-Graduate Student(s) working with the Supervisor at Present:

Md. Ruhul Amin ----------	**Names and signatures of the members of the Doctoral Committee (if applicable)**	
Signature of the Student	1.	
12.11.23	2.	
----------	3.	
Signature of the Supervisor	4.	
	5.	
	6.	
	7.	
----------	8.	
Signature of the Co-Supervisor		

Signature of the Head of the Department		

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Sl. No 31

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Rajshahi University of Engineering & Technology

RUET

GRADE SHEET

POST GRADUATE ODD SEMESTER EXAMINATION, 2023
Department of Computer Science & Engineering

Name of the Student: **Md. Ruhul Amin**

Roll No.: **2204103014** Registration No.: **74/2022-2023** Session: **2022-2023**

Course No.	Course Title	Credit	Grade
CSE 6106	Combinatorial Optimization	3.00	A+
CSE 6310	Data Mining and Warehousing	3.00	A

Credit Registered	Credit Earned	Cumulative Earned Credit	GPA	CGPA
6.00	6.00	6.00	3.75	3.75

Prepared by **Checked by** **Controller of Examinations**

Attested

ডাঃ মোঃ ইমরুল ইসলাম (বিসিএস-১২৬১৮৬)
এমবিবিএস, এমডি-এনেসথেসিয়া (বিএমএএ২৫৪)
সিনিয় স্কেল (এনেসথেসিয়া), প্রভাষক, ডিপ্লিইন্টেন্স, রাবেকা, রাজা
রাজেন্দ্র সিনিয়র কনসালটেন্ট (এনেসথেসিয়া)
রাজশাহী মেডিকেল কলেজ হাসপাতাল, রাজশাহী।

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September 03, 2024

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Sl. No ..14.....

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Rajshahi University of Engineering & Technology

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY * BANGLADESH

RUET

GRADE SHEET

POST GRADUATE EVEN SEMESTER EXAMINATION, 2023
Department of Computer Science & Engineering

Name of the Student: **Md. Ruhul Amin**

Roll No.: **2204103014** Registration No.: **74/2022-2023** Session: **2022-2023**

Course No.	Course Title	Credit	Grade
CSE 6501	Advanced Digital Image Processing	3.00	A
CSE 6701	Advanced Database Management System	3.00	A+

Credit Registered	Credit Earned	Cumulative Earned Credit	GPA	CGPA
6.00	6.00	12.00	3.75	3.75

Prepared by **Checked by** **Controller of Examinations**

ডাঃ মোঃ ইমরুল ইউসুফ (বিএসএস(৪২৪১৮৬)
এমবিবিএস, এমডি-এনেসথেসিয়া (বিএসএমএমইউ)
সিনিয়র কন্সালটেন্ট (এনেসথেসিয়া), ওএসডি, ডিজিএইচএস, মহাখালী, ঢাকা
সাবেক সিনিয়র কনসালটেন্ট (এনেসথেসিয়া)
রাজশাহী মেডিকেল কলেজ হাসপাতাল, রাজশাহী।

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December 17, 2024

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Sl. No ....04....

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Rajshahi University of Engineering & Technology

GRADE SHEET

POST GRADUATE EVEN SEMESTER EXAMINATION, 2024
Department of Computer Science & Engineering

Name of the Student: **Md. Ruhul Amin**

Roll No.: **2204103014** Registration No.: **74/2022-2023** Session: **2022-2023**

Course No.	Course Title	Credit	Grade
CSE 6303	Machine Learning	3.00	A+
CSE 6602	Advanced Computer Networks	3.00	A

Credit Registered	Credit Earned	Cumulative Earned Credit	GPA	CGPA
6.00	6.00	18.00	3.75	3.75

Prepared by **Checked by** **Controller of Examinations**

Attested
28/7/25
ডাঃ মোঃ ইমরুল ইসলাম (বিসিএস-১২৪১৮৬)
এমবিবিএস, এমডি-এনেসথেসিয়া (বিএসএমএমইউ)
সিনিয়র কনসালটেন্ট (এনেসথেসিয়া), ওএসডি, ডিজিএইচএস, মহাখালী, ঢাকা
সংযুক্তঃ সিনিয়র কনসালটেন্ট (এনেসথেসিয়া)
রাজশাহী মেডিকেল কলেজ হাসপাতাল, রাজশাহী।

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May 06, 2025

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